4. Overview of Quantitative Trading

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1. Irrational Behaviors

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1. Irrational Behaviors in Trading: Behavioral Features of Different Roles

	Quantitative Investor	Entrepreneur/Professional/Scientist	Gambler/Speculator
Perception of Risk	Risk is a quantifiable and manageable probability problem.	Risk is the cost of innovation and opportunity; "I am the master of my life."	Praying to gods and relying on luck.
Decision-Making Basis	• Data models • Probabilistic thinking • Strict stop-loss rules	• Enthusiasm • Rational Validation • Resource integration capability • Long-term trend judgment	• Intuition/emotional impulses • "Hot-hand fallacy" (superstition of winning streaks) • Irrational wishful thinking
Primary Motivation	Capital appreciation	Value creation (solving real-world problems + building sustainable inventions/business models)	Instant gratification (dopamine rush + short-term excitement from winning)
Attribution of Failure	• Strategy failure • Risk control errors • Market black swan events	• Product-market misfit • Limited time/fundings • Inapproperate approach/management	• "Bad luck" • Perceived dealer manipulation
Time Perspective	Short-to-medium (from minutes to months)	Long-term (from years to decades)	Immediate (from single bets to hours)
Source of Psychological Resilience	Discipline (adhering to trading plans)	Sense of mission	Addiction (inability to stop betting)

The difference between a trader and a gambler is the rationality versus irrationality of their decision-making process.
The distinction between a trader and an entrepreneur is the length and focus of their time horizon.

1. Irrational Behaviors in Trading: Partial List of Bias and Mistakes in Trading

Cognitive Bias
- Confirmation Bias
- Loss Aversion
- Recency Bias
- Anchoring Bias
- Herd Mentality Bias
- Ambiguity Aversion

Emotional Bias
- Fear
- Impatience
- Greed
- Over confidence
- Desperation
- Regret
- Hope

Biased Reaction
- Information Overload

News, rumors, opinions, reports, policies, economics, etc.

- Flip-flop

Tesla is legendary! -> Tesla gonna go bankrupt!
Wait a miniute! I should/shouldn't...

- Early/Late Entry, Early/Late Exit

Close winners early and let losers run.
Too greedy to exit at the best time.
Can't wait to open a trade.

- Poor Risk Management

Setting stoploss/stopgain too small/large without backtests
Leverage at a wrong time/level
Under/Over Diversification

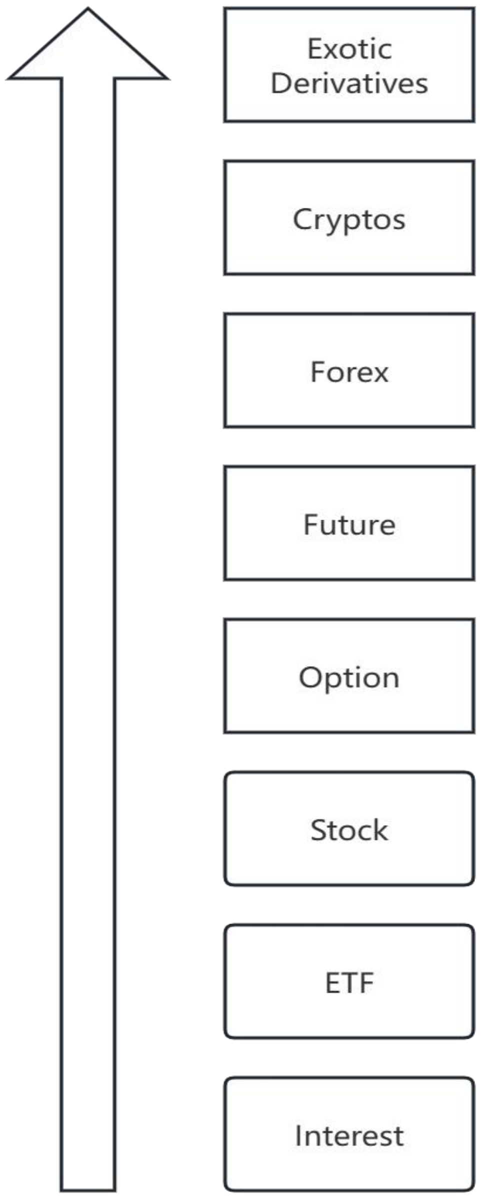
https://www.morganstanley.com/cs/pdf/619598-3174306-MSVA-Behavioral-Guide-r7.pdf
Statman, M. (2019). Behavioral finance: The second generation. CFA Institute Research Foundation.

1. Irrational Behaviors in Trading: Behavioral Features of Different Roles

Forbes:
10% of day traders actually make money.

Tradeciety :
40% of all futures day traders quit in 4 months,
80% quit within a year, only 7% are able to last 5 years or more.
Among the 20% who last over a year, not all of them are profitable, just persistent.

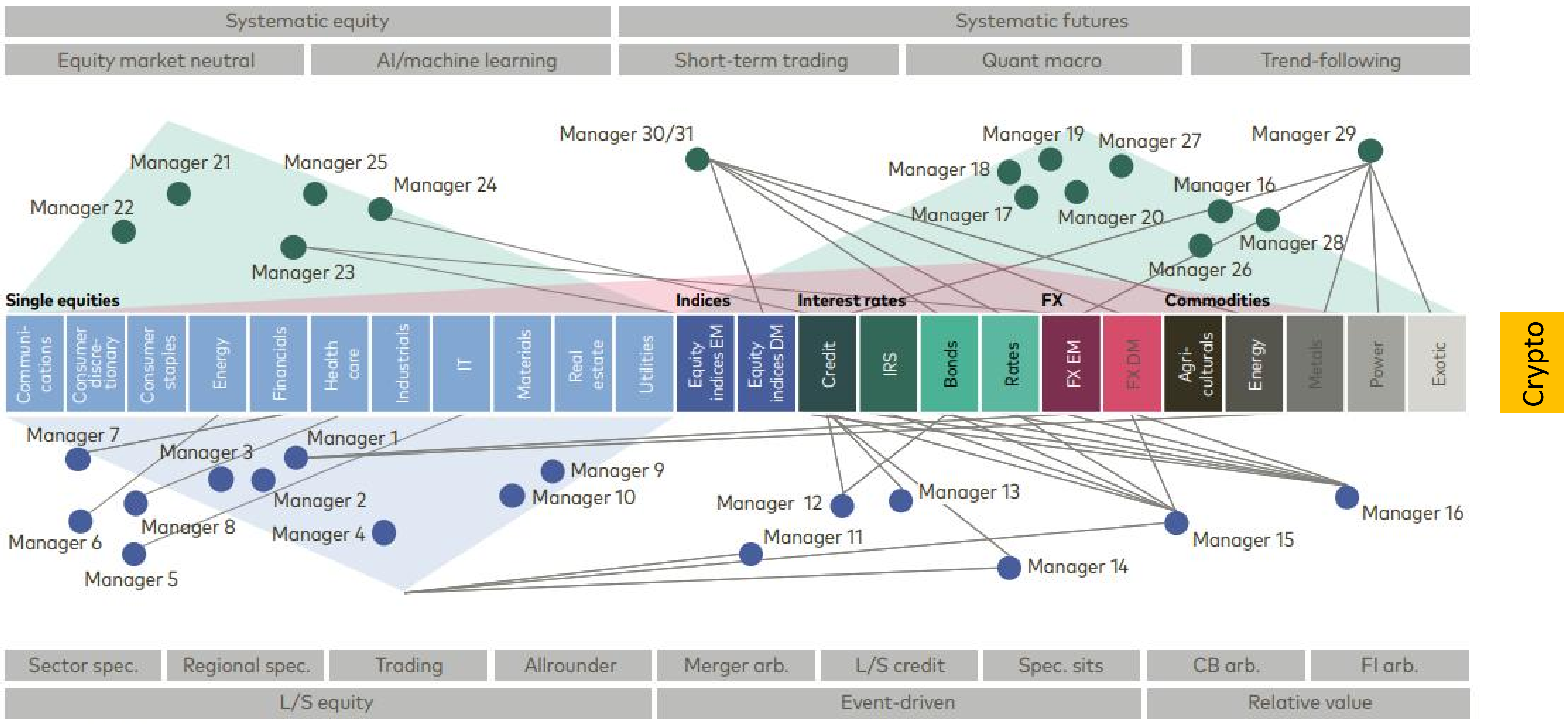
Cryptos are more challanging than any other markets!
If you could survive in crypto trading, you'll find the stock market is easy and boring.



Levels of Difficulty

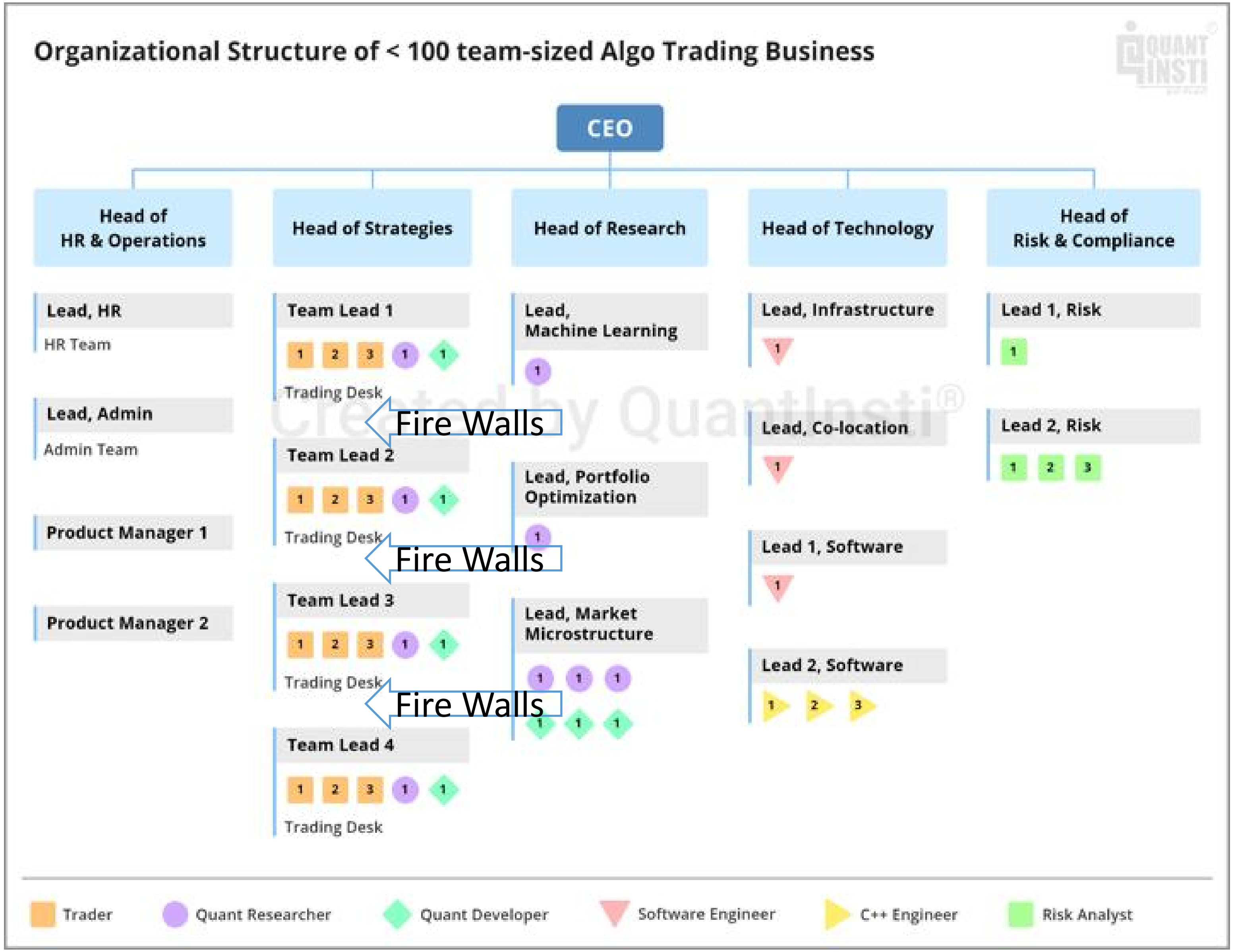
2. Overview of Quantitative Trading

2. Overview of Quantitative Trading: Types of Quant Strategies



Source: LGT Capital Partners

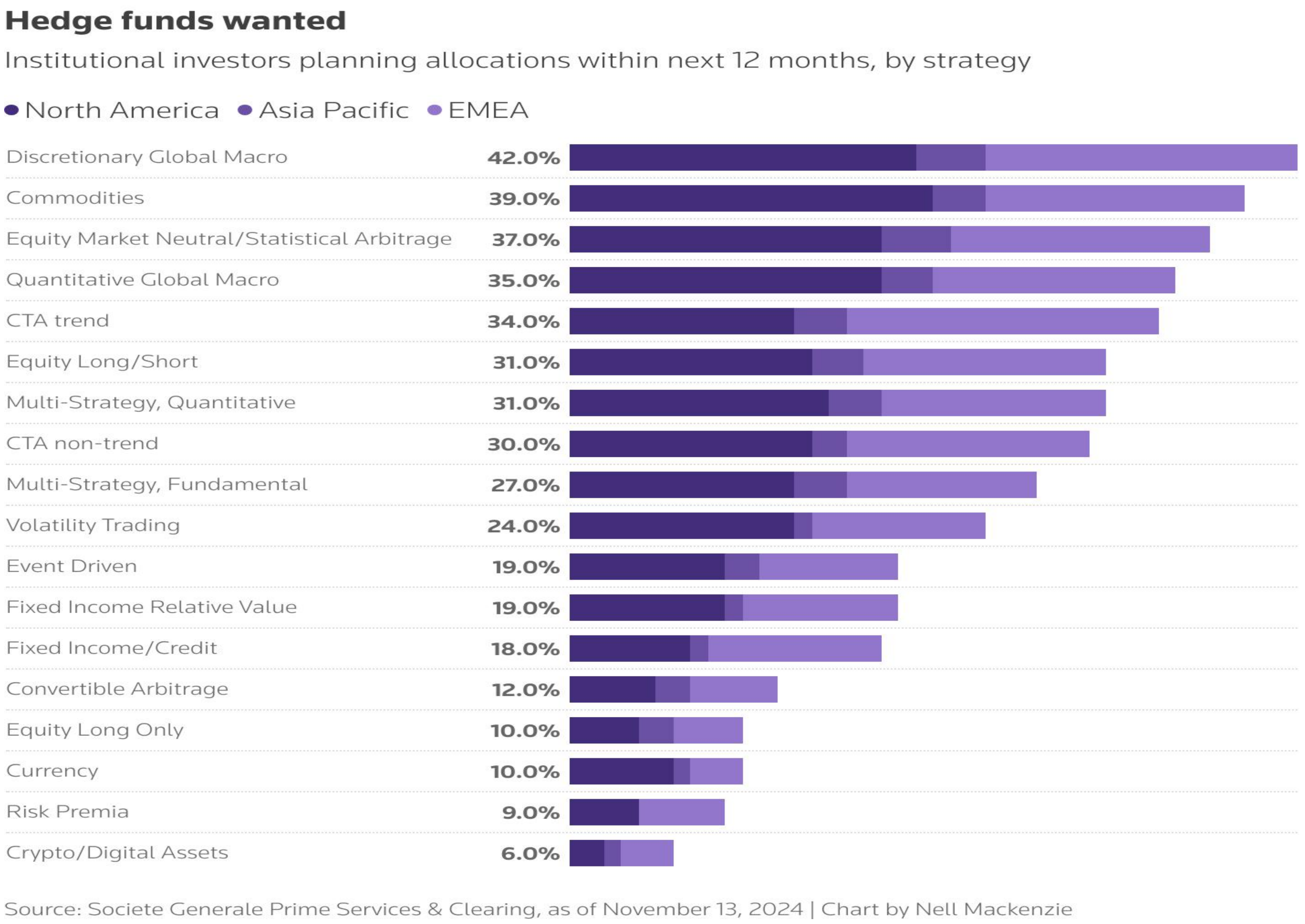
2. Overview of Quantitative Trading: Roles in a Quant Firm



Type	Average Annual Compensation
Quant Trader	$316,764
Quant Researcher	$173,119
Quant Developer	$110,000

source:wallstreetoasis.com

2. Overview of Quantitative Trading: Trending in Quant Strategies



3. Quantitative Trading Strategy Design

3. Quantitative Trading Strategy Design: Core Principles of Quantitative Trading

- **Data-Driven**

Human: “Close all the positions when the market is in panic.”

Quant: “If VIX > 20, close leveraged positions. If VIX>40, close 100% of long side equity positions.”

- **Explicit, Rule-Based Design**

Human: “Seems like it’s starting to drop, so I’ll close out everything in the next few days.”

Quant: “Mandatory liquidation conditions: Indicator_X<0 & sum(r(i-7:i)<0)>5.”

- **Measurable Outcomes**

Human: “Modify the strategy if it underperforms.”

Quant: “If a strategy's loss reaches 5%, then transfer it to the observation group and reduce its portfolio weight to 50%. If a strategy's loss reaches 10%, then transfer it to the warning group, reduce its weight to 0%, but continue performance monitoring. If a strategy's loss reaches 15%, then transfer it to the recycle group and suspend all further tracking.”

3. Quantitative Trading Strategy Design: Core Principles of Quantitative Trading

- **Statistical Rigor**

Human: “Setting T=5 and Stoploss=5%, the strategy looks good.”

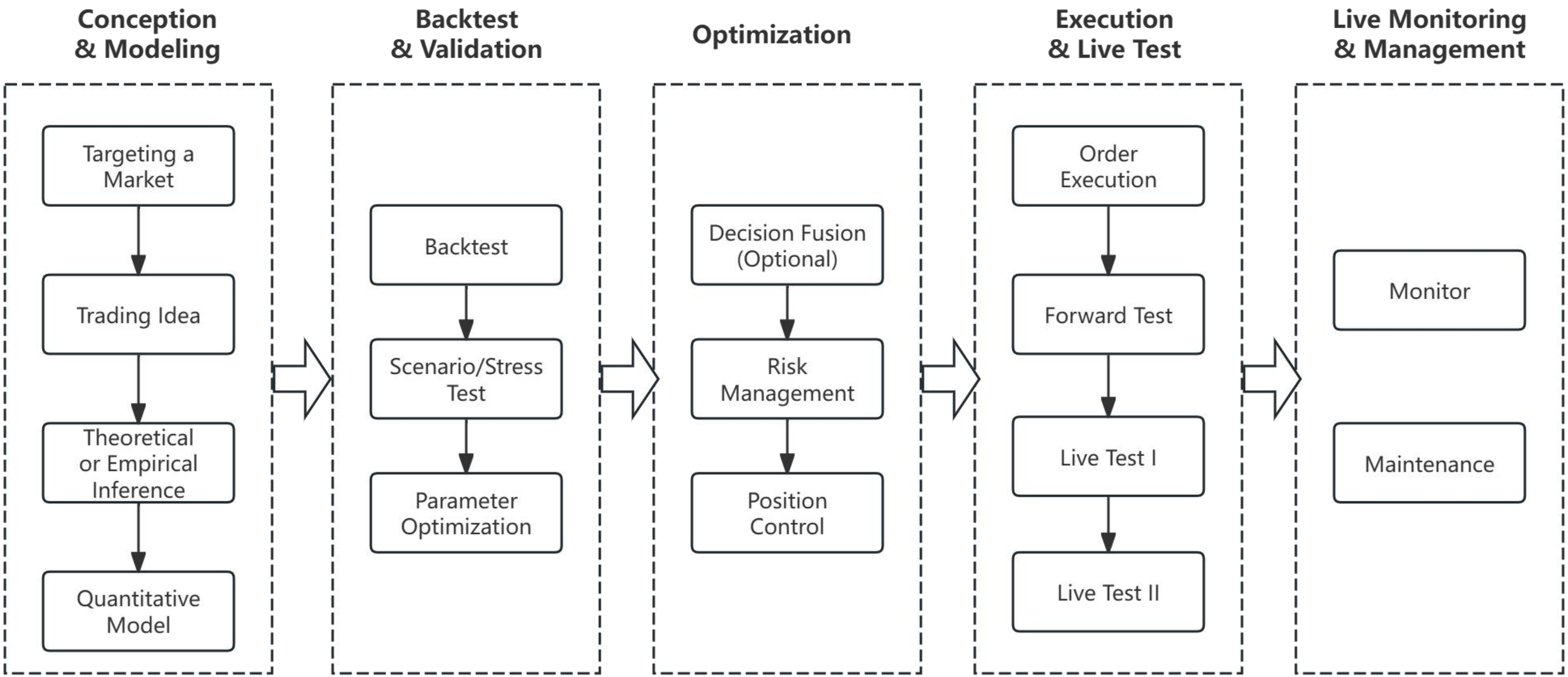
Quant: “In-sample backtesting from 2020 to 2024 with a parameter set of T=5 yields the strategy Sharpe Ratio of 1.5 and a maximum drawdown of 24.5%. The introduction of a 5% stoploss enhances risk-adjusted returns, raising the Sharpe Ratio to 1.75 and curtailing the maximum drawdown to 14.5%. However, this risk mitigation comes at the expense of a 5% decline in total return.”

- **Unconditional Execution**

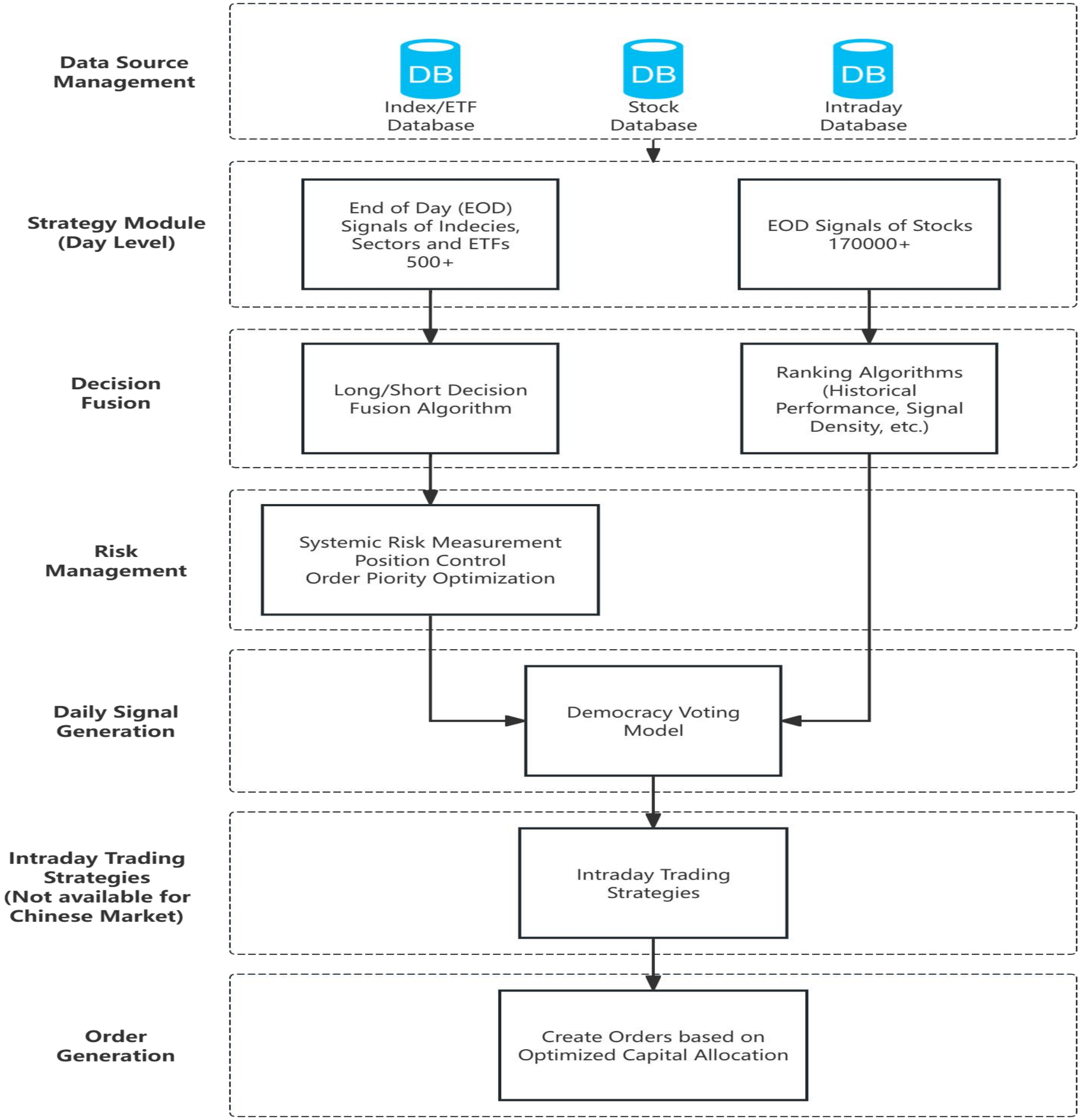
Human: “I follow the signals, but there are moments when manual control can enhance performance.”

Quant: “When open a position, set a limit order at yesterday’s close price and send it 15 mins after the market opening. If it is not executed by 14:30 pm, cancel the order; If it is partially executed, record position changes and cancel the rest pending amount.”

3. Quantitative Trading Strategy Design: Quant Strategy Development Lifecycle



3. Quantitative Trading Strategy Design: Components of My Own Trading System



- A simple quant strategy can be as simple as a moving average.
- A quantitative trading system serves as an automated representation of a quant team's functions, including data analysis, trading decision making, risk management, portfolio management, and order management, inventory management, etc.
- The system may be very different acording to the strategies.

3. Quantitative Trading Strategy Design: How to Observe the Market? (Idea)

Phenomenon Category	Primary Driver	Examples & Concepts	Trading/Investment Implication
Market Anomalies	Persistent Return Predictability	Size, Value, Momentum. Volatility anomaly Post-Earnings Announcement Drift	“Strong” anomalies could be the triggers to a new trend or pattern. “Weak” anomalies will disappear soon and the market will go back to normal.
Price, Volume, Volatility Patterns	Recurring Price & Chart Behavior	Day-of-week Effect Overnight Drift “U” Shape of Intraday Volume	Identifying probable entry/exit points, directions, based on the premise that "history tends to repeat itself".
Behavioral & Sentiment Phenomena	Systematic Investor Psychology	Herding Behavior: Bubbles, Panics. Overreaction/Underreaction. Sentiment Indicators: Social Listening.	Contrarian or momentum strategies based on crowd sentiment extremes.
Structural & Microstructural Phenomena	Market Mechanics & Frictions	Order Flow Imbalances Put-Call Ratio Near-far Future Spread	Directional assessment, relative value
Fundamental Mispricing	Mispricing vs. Intrinsic Value	Violations of Law of One Price. Piotroski F-Score.	Value investing, cross-exchange arbitrage, accounting-based strategies.

3. Quantitative Trading Strategy Design: Underlying Patterns of Quantitative Models

Economic Pattern

- Fundamental Analysis
- Macro Strategy
- Supply Chain Arbitrage
- Portfolio Theory

Pros:
Solid in logic, low-turnover asset allocation (easy to execute), persistent
Cons:
May not be replicable (e.g. 2008 Crisis)

Statistical Pattern

- Factor Models
- Technical Inidcators
- Statisitcal Arbitrage
- AI

Pros:
Testable, capturing more opportunities
Cons:
May be statistically right, but logically wrong; vulnerable to non-stationarity.

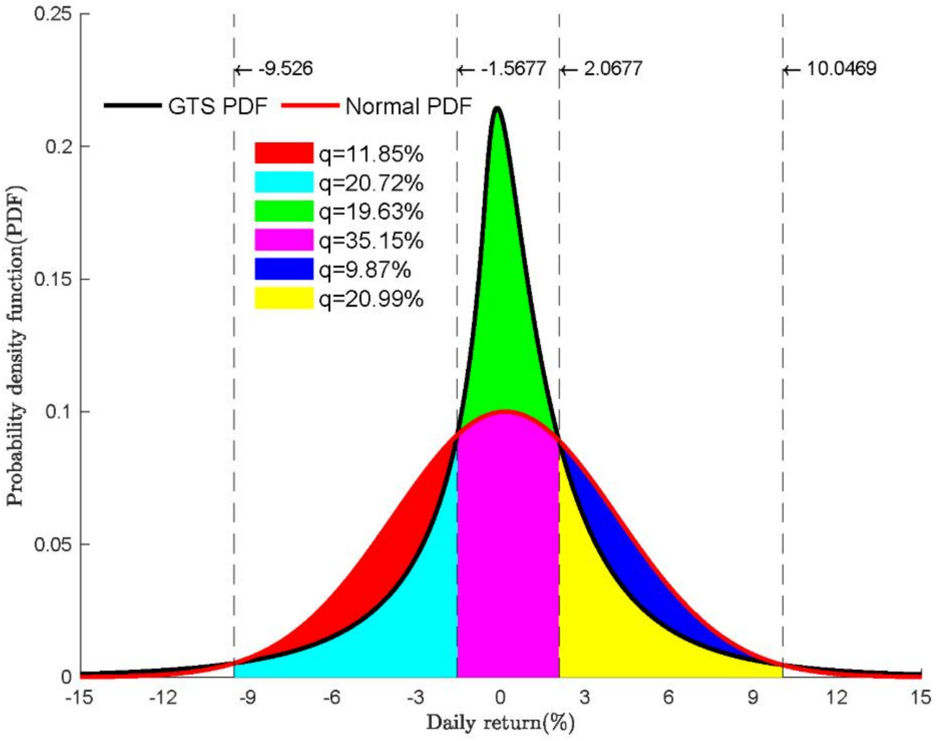
Scientific Pattern

- Heat Equation (Black-Scholes model)
- Network Theory in Systemic Risk Measurement
- Chaos Theory
- Classical Mechanics

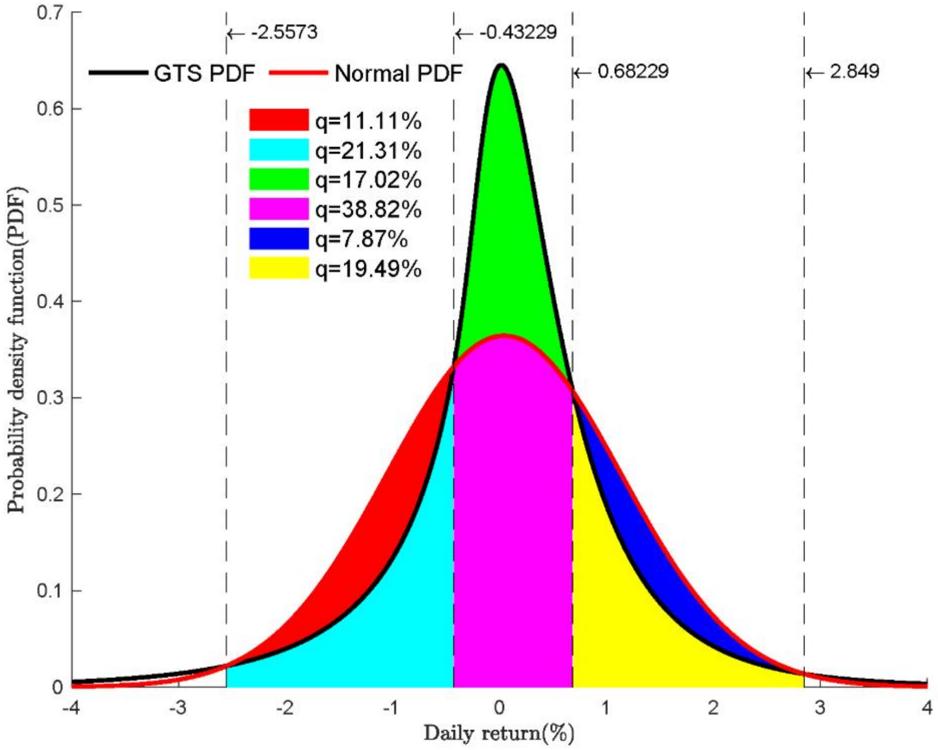
Pros:
Testable, persistent, strong in logic.
Cons:
Vulnerable to non-stationarity, difficult for modeling and mapping.

4. Stock vs. Crypto

4. Stock vs. Crypto: Major differences between crypto market and stock market



(**a**) Bitcoin returns distribution.



(**b**) S&P 500 returns distribution

source: Nzokem, A., & Maposa, D. (2024)

- Crypto market is highly volatile/risky with big fat tails.

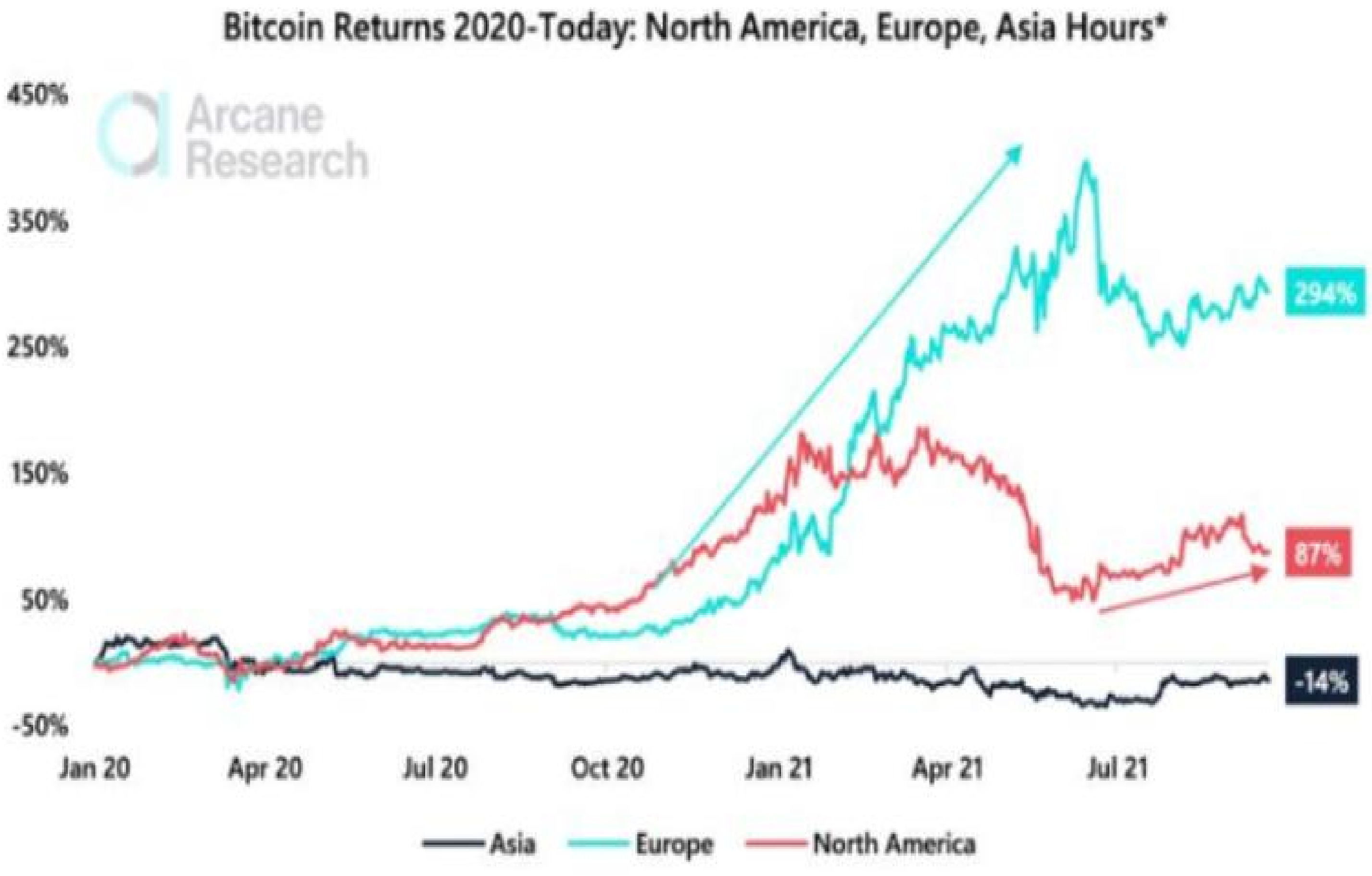
4. Stock vs. Crypto: Major differences between crypto market and stock market

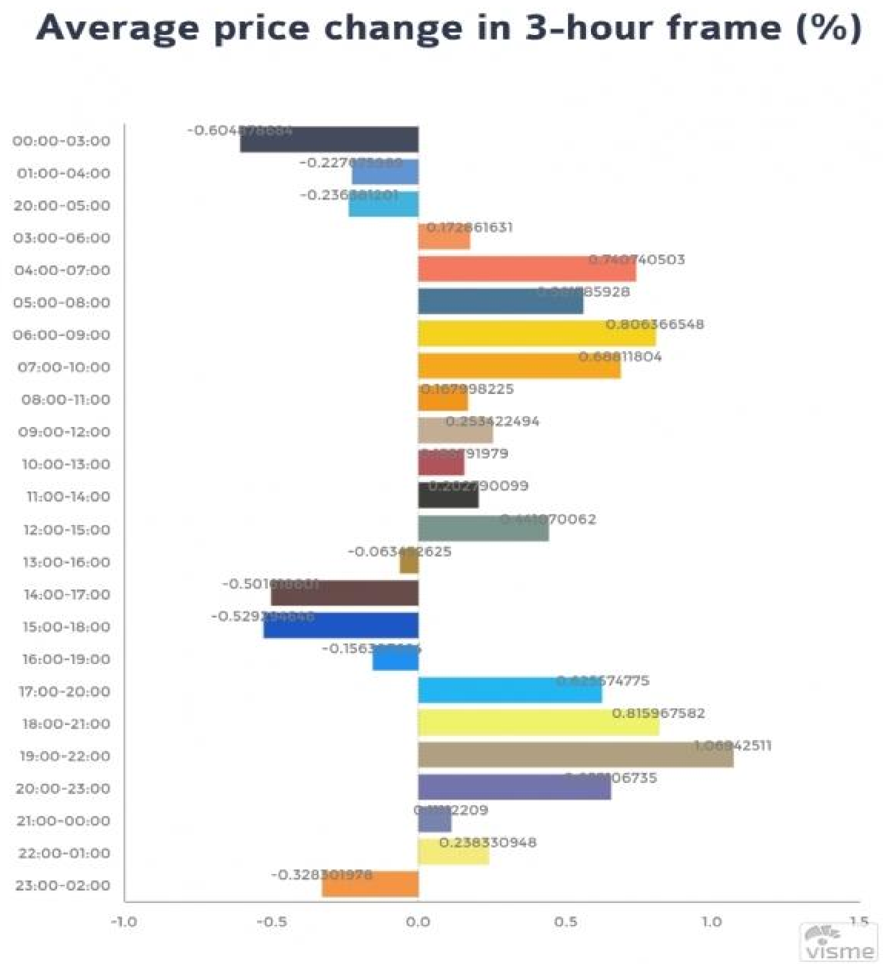
source: Bloomberg

source: bitbo.io

- Normal: Volatility(BTC) > Volatility(SPY) vs Abnormal： Volatility(BTC) < Volatility(SPY)
- Short SPY, long BTC; Short VXX, long BTC Straddle.

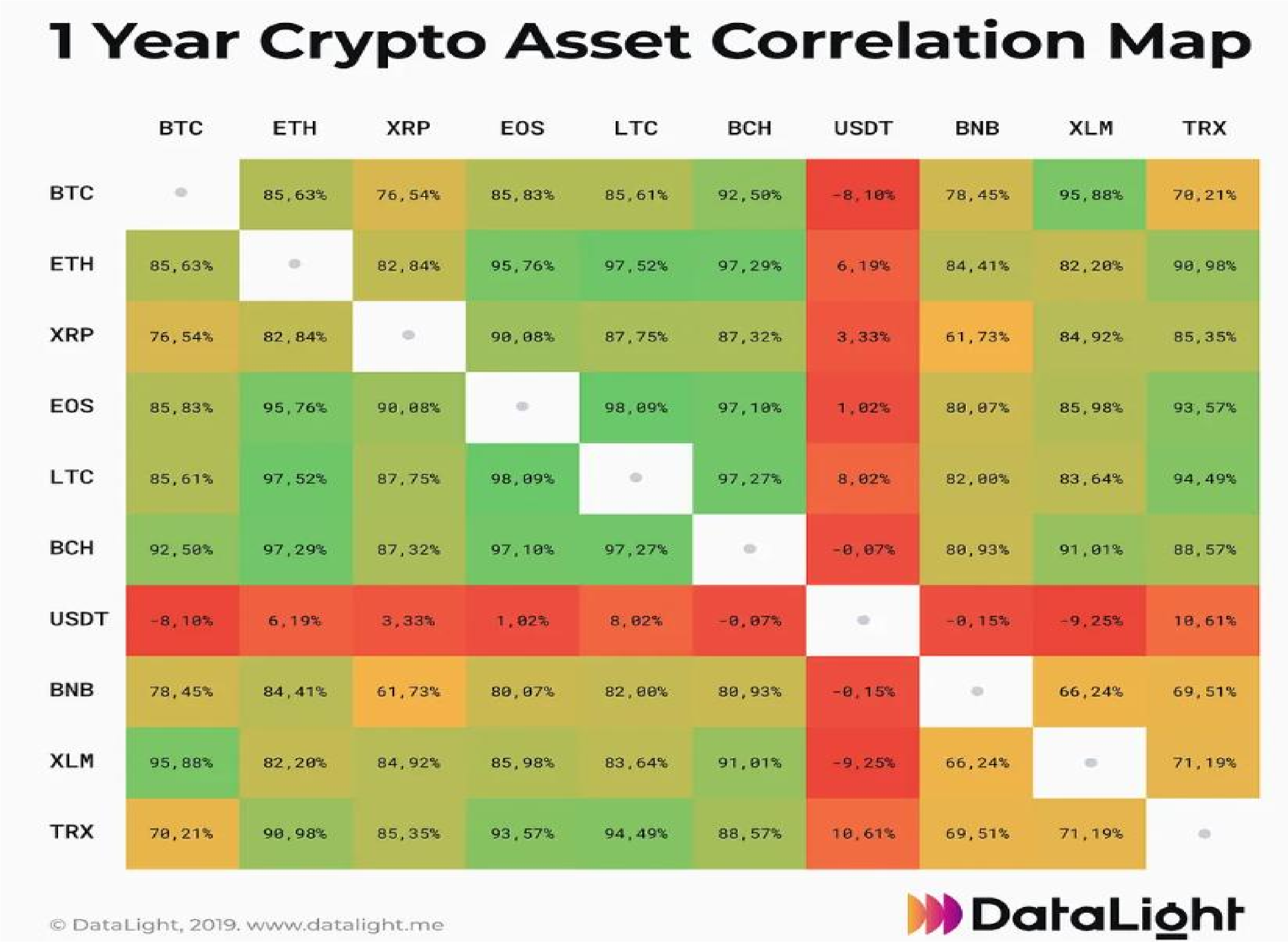
4. Stock vs. Crypto: Major differences between crypto market and stock market





- Crypto market is a 24-7 market: US time, Asia time, Eurpoe time may behave different.

4. Stock vs. Crypto: Major differences between crypto market and stock market



1 Year Crypto Asset Correlation Map

	BTC	ETH	XRP	EOS	LTC	BCH	USDT	BNB	XLM	TRX
BTC	•	85,63%	76,54%	85,83%	85,61%	92,50%	-8,10%	78,45%	95,88%	70,21%
ETH	85,63%	•	82,84%	95,76%	97,52%	97,29%	6,19%	84,41%	82,20%	90,98%
XRP	76,54%	82,84%	•	90,08%	87,75%	87,32%	3,33%	61,73%	84,92%	85,35%
EOS	85,83%	95,76%	90,08%	•	98,09%	97,10%	1,02%	80,07%	85,98%	93,57%
LTC	85,61%	97,52%	87,75%	98,09%	•	97,27%	8,02%	82,00%	83,64%	94,49%
BCH	92,50%	97,29%	87,32%	97,10%	97,27%	•	-0,07%	80,93%	91,01%	88,57%
USDT	-8,10%	6,19%	3,33%	1,02%	8,02%	-0,07%	•	-0,15%	-9,25%	10,61%
BNB	78,45%	84,41%	61,73%	80,07%	82,00%	80,93%	-0,15%	•	66,24%	69,51%
XLM	95,88%	82,20%	84,92%	85,98%	83,64%	91,01%	-9,25%	66,24%	•	71,19%
TRX	70,21%	90,98%	85,35%	93,57%	94,49%	88,57%	10,61%	69,51%	71,19%	•

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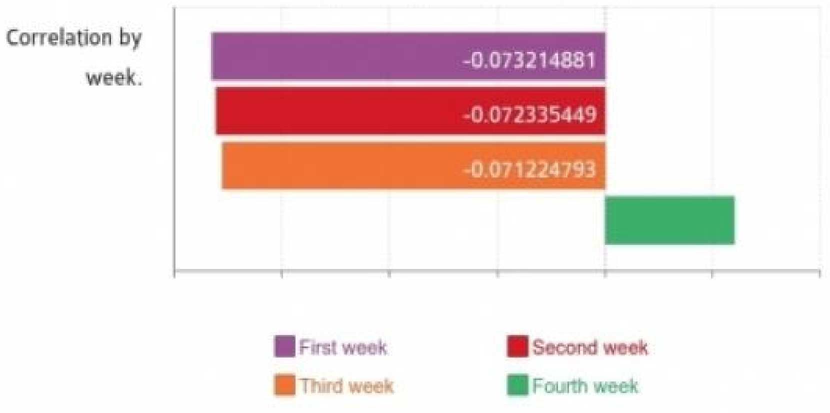

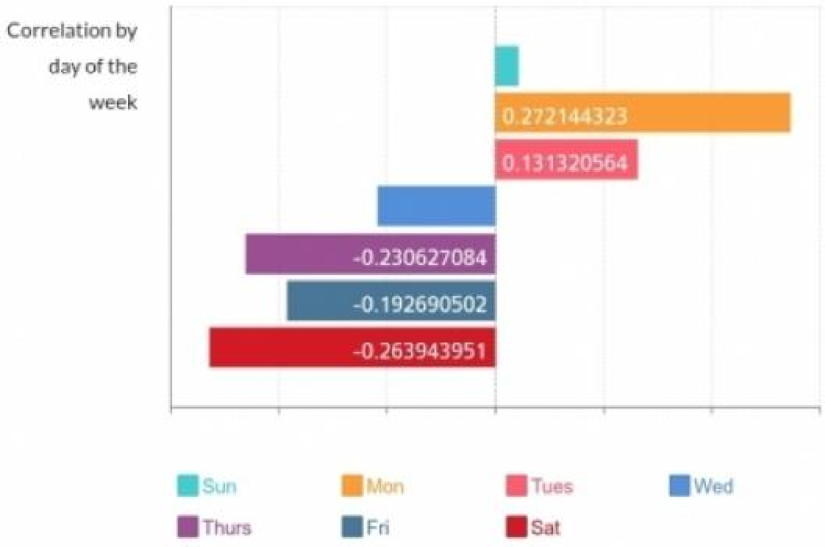

source: totalcoin.io

- Crypto market has a very high correlation that all diversification in risk management may fail!
- Relative value strategies MAY work better during the fourth week of a month/Mon, Tues of a week.

Assignment 1 (Regular Track): Crypto Market Analysis

Lecture 4

1.(30) **NO AI HERE!** Try to practice on your own mind and finish it before Lecture 5. That’s a daily routine for quant researchers and traders.

Write an essay (200-500 words) about your trading idea. Illustrate your logic, pattern, model, trading rules and expected outcomes. This is the very first strategy draft: no detailed data, testing, or code is required.

Lecture 5

2.(70) **AI is fine here.** Reorganize your trading idea with Table 1 (Core Logic Design of Trading Strategies).

Originality and novelty in your trading idea is encouraged and will be considered in grading. However, note that a clear, well-reasoned, and critical analysis of an existing concept can be just as valuable as a completely novel idea.

Resources/Events:

For PhD students

- Citadel, GQS PhD Colloquium

https://www.citadel.com/careers/programs-and-events/gqs-phd-colloquium/

- Bloomberg, PhD Fellowship

https://www.bloomberg.com/company/values/tech-at-bloomberg/quantitative-finance-phd-fellowship/

For everyone,

- Citadel, Competition

https://www.citadel.com/careers/programs-and-events/terminal/

- CME, Competition

https://www.cmegroup.com/events/university-trading-challenge.html

- Kaggle, Competition

https://www.kaggle.com/competitions/hull-tactical-market-prediction

- Jane Street

Strategy and Product Talk at Duke, 04 February

Thank You

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